

Partial Differential Equation

Reference book: Ordinary and Partial Differential Equations: M. D. Raisinghanian

Part-III, Chapter name: Origin of partial Differential Equations

Definition: An equation containing one or more partial derivatives of an unknown function of two or more independent variables is known as PDE.

Example: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0$

Notation: Let z be a function of two variables x and y .

So, $p = \frac{\partial z}{\partial x}$, $q = \frac{\partial z}{\partial y}$, $r = \frac{\partial^2 z}{\partial x^2}$, $s = \frac{\partial^2 z}{\partial x \partial y}$, $t = \frac{\partial^2 z}{\partial y^2}$

Suffix Notation: $z_x = \frac{\partial z}{\partial x}$, $z_y = \frac{\partial z}{\partial y}$, $z_{xx} = \frac{\partial^2 z}{\partial x^2}$, $z_{xy} = \frac{\partial^2 z}{\partial x \partial y}$, $z_{yy} = \frac{\partial^2 z}{\partial y^2}$

Origin of PDE: PDEs can be formed by the elimination of arbitrary constants or arbitrary functions.

Example 1: Elimination of arbitrary constants from the following equations and form corresponding partial differential equations.

- a) $z = Ae^{pt} \sin px$
- b) $z = (x - a)^2 + (y - b)^2$
- c) $z^2 + (x - h)^2 + (y - k)^2 = \gamma^2$
- d) $z = axe^y + \frac{1}{2}a^2e^y + b$
- e) $z = ax + by + a^2 + b^2$
- f) $az + b = a^2x + y$

Example 2: Eliminate the arbitrary function ϕ from the equation $\phi(u, v) = 0$ (where u and v are function of x, y and z) and form corresponding partial differential equations.

- a) $\phi(x + y + z, x^2 + y^2 - z^2) = 0$
- b) $\phi(x + y + z, x^2 + y^2 - z^2) = 0$
- c) $y = F(x - at) + G(x + at)$

Exercise 1(A): 1- 6

Exercise 1(B): 1-4, 8

Solution of Partial Differential Equation

Lagrange's Method:

Lagrange's equation is a one kind of partial differential equation.

The general form of the Lagrange's equation is:

$$Pp + Qq = R;$$

where P, Q and R are functions of x, y, z .

Example: $xyp + yzq = zx$

Theorem: The general solution of Lagrange's equation is $\phi(u, v) = 0$;

where, ϕ is an arbitrary function and

$$u(x, y, z) = c_1 \quad \text{and} \quad v(x, y, z) = c_2$$

are two independent solutions of

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

Here, c_1 and c_2 are arbitrary constants and **at least one** of u and v must contain the z variable.

Examples: Solve the following Lagrange's equations

1. $\frac{y^2z}{x}p + xzq = y^2$
2. $y^2p - xyq = x(z - 2y)$
3. $xzp + yzq = xy$
4. $(mz - ny)p + (nx - lz)q = ly - mx$
5. $x(y - z)p + y(z - x)q = z(x - y)$
6. $(y + z)p + (z + x)q = x + y$